Utsav Amrish Joshi

Mechanical Engineer - Manufacturing & Process Engineering

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Manufacturing engineer with plant-floor experience in high-volume food processing and a simulation-heavy master's background. Work spans time-study and capacity analysis, automation feasibility and ROI, designed experiments, and coupled thermal modeling - measuring what a process actually does, then quantifying what changing it is worth.

Education

M.S. Mechanical Engineering - Arizona State University, USA — May 2026 | GPA 3.43

B.E. Mechanical Engineering - Gujarat Technological University, India — June 2023 | GPA 3.33

Manufacturing Engineering Internship - Shamrock Foods Company

Gold Canyon Meat Co. | May - July 2026 | ~450,000 sq ft meat-processing plant, 11 production lines | Reported to the Enterprise Maintenance Manager

Automated Sorting & Labeling System - Time Study and Automation ROI

Time study · labor-cost modeling · capital justification | Excel

Answered a single question for plant leadership: how much warehouse labor is spent per pound and per case today, and how much of it could a conveyor-based sorting and labeling system realistically remove? Scoped the analysis to the same-day JIT/STD shipping workflow - the most labor-intensive and the best conveyor fit - on the reasoning that a system able to handle it would absorb the remaining make-to-stock volume with room to spare.

- **Measured, not assumed:** ran floor time studies to derive labor-minutes per case for each of four workflow types, then applied them to the daily case mix (2,551 cases/day at a 21 lb weighted-average case) to isolate 48.0 labor-hr/day - 30% of all warehouse labor - as automation-addressable.
- **Netted out what automation won't remove:** built a residual manual-workload model (monitoring, staging, sensor and scale checks, waste removal, maintenance) totaling 25.4 labor-hr/day, leaving 22.6 hr/day genuinely automatable rather than reporting the gross figure.
- **Discounted the savings to what's actually capturable:** labor cost only converts to cash when paid hours are cut, so applied a reduced 15% overhead factor and modeled capture-rate scenarios by whole-FTE increments instead of quoting a single idealized number.
- **Recommendation:** against a $3.0M installed cost, ROI ran 5.9% (16.8-yr payback) at current throughput and 9.96% (10.0-yr payback) at a projected +25% throughput - but 15-year NPV stayed negative at an 8% hurdle rate in both cases, supporting a documented “not at this throughput” recommendation rather than a favorable number reverse-engineered to justify the spend.

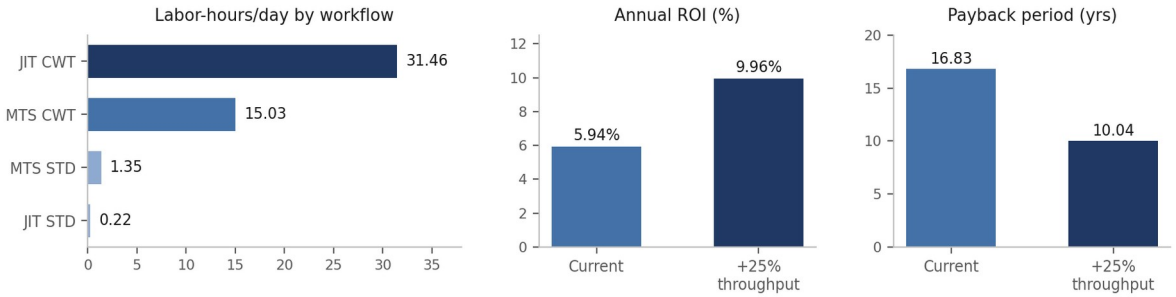


Fig. 1 - Time-study labor distribution by workflow type, and resulting ROI / payback under both throughput scenarios

Equipment Capacity Analysis

Capacity utilization · bottleneck identification | Excel

Benchmarked actual throughput against rated design capacity across 11 production lines and 30+ machines, building a consolidated workbook of machine specifications, rated capacity, and observed operating capacity - including a +25% throughput scenario to align with the automation study.

- **Triangulated three data sources:** floor time studies of machines in operation, each machine's OEM design data book, and direct operator interviews - rather than relying on nameplate figures alone.
- **Finding:** most lines ran materially below rated capacity, traced primarily to frequent product changeovers (from a few hours to every other day), flagging changeover frequency rather than machine performance as the constraint worth attacking.

Production Floor Rebuild - Phased Equipment Reallocation

Layout planning · production continuity | AutoCAD, SolidWorks

Planned equipment relocation for a full production-floor replacement - the original slab was not meat-plant grade - with the binding constraint that production could not stop. Sequenced the rebuild into five zone-based phases.

- **Each phase had to preserve five separate flows simultaneously:** worker entry/exit, forklift movement, finished-goods out, raw-goods in, and maintenance access.
- **Hardest constraints:** one machine that could not be powered down at any point in the sequence, and a traffic chokepoint that had to stay open throughout.

Dairy QA Knowledge Base Optimization

Process improvement · deployed tool | Microsoft Copilot Studio, SharePoint, Teams

QA technicians and supervisors were losing significant time hunting for SOPs and regulatory documentation (FDA, USDA, PMO, SQF, HACCP/HARPC) scattered across systems - during audits, line checks, and corrective-action investigations, where delay and stale information both carry real risk. Built and deployed an AI agent in Copilot Studio over a structured SharePoint knowledge base so the team could ask in plain language and get either a direct answer or the document's location.

- **Result:** cut document search time from roughly 30 minutes to about 1 minute.
- **Adoption:** tested and used by the full QA management and technician group.
- **Why it mattered:** returned experienced QA staff to hands-on quality oversight and reduced the chance of acting on outdated documentation during an audit.

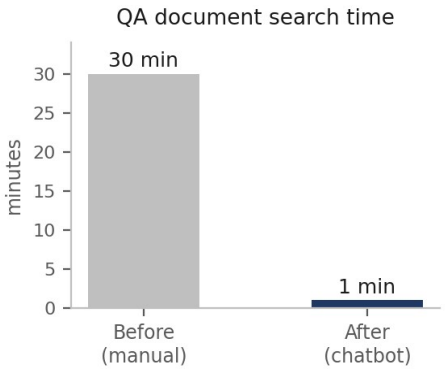


Fig. 2 - QA document search time, before vs. after

Graduate Research & Coursework - Arizona State University

Design of Engineering Experiments - Factorial Study of Cooling Performance

IEE 572 | JMP

Designed and executed a full 2^3 factorial experiment (8 combinations × 2 replicates = 16 randomized runs) to quantify how ice geometry, ice mass, and carbonation drive cooling rate and temperature retention - with nuisance and controlled factors explicitly identified and held before any data was taken.

- **Individual contribution:** personally designed the experiment, ran all 16 trials, and performed the full statistical analysis and write-up.
- **Analysis:** half-normal plots to screen effects, ANOVA for significance, regression modeling, residual diagnostics, and response-surface plots.
- **Cooling rate:** ice quantity dominated at ~95% of variability ($p<.0001$); ice shape was smaller but real at ~3% ($p=.0212$); carbonation was not significant. Model $R^2 = 0.97$.
- **Retention time:** ice quantity again dominant (~77%, $p<.0001$), with ice shape and the carbonation * shape interaction each ~9% and significant. Model $R^2 = 0.94$.
- **Closed the loop:** reduced the fitted models to operating recommendations rather than stopping at the statistics.

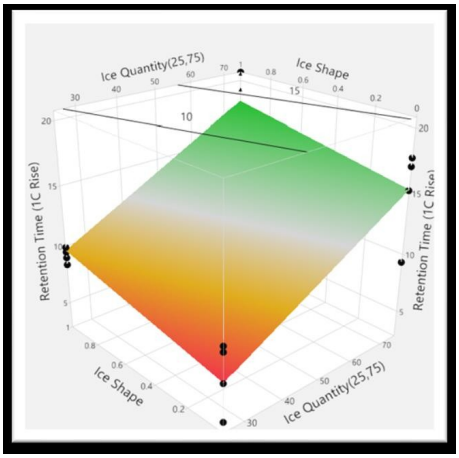


Fig. 3 - Response surface: retention time vs. ice quantity and shape

Coupled Electromagnetic-Thermal Modeling of Microwave Heating

Multiphysics simulation | COMSOL

Built a frequency-domain Maxwell solve coupled to a transient heat-transfer model (2.45 GHz cavity, 250 mL water load), using cycle-averaged resistive loss to bridge the two timescales - the standard approach for any microwave-driven industrial heating, curing, or drying process.

- **Validated before trusting:** hand-calculated penetration depth and expected temperature rise first, then checked the solved model against those figures and against an energy/power balance - treating a simulation result as a claim requiring verification.
- **Physics captured:** temperature-dependent permittivity via a Debye relaxation model, reproducing the self-stabilizing feedback where absorption falls as the load heats, and the resulting non-uniform temperature distribution.

Thermal Simulation of Additive Manufacturing Processes

Two FEA studies, Jan - Apr 2025 | ANSYS, COMSOL

Two transient thermal studies of heat input and heat-affected zones in additive processes, chosen to compare how material properties and scan strategy each govern thermal history.

- **FDM (ANSYS):** modeled layer-by-layer heat buildup using the element activation method, comparing ABS and PA12. ABS retained more heat and produced a larger heat-affected zone, consistent with its higher glass-transition temperature; PA12 cooled faster with better thermal stability.
- **Laser powder bed (COMSOL):** modeled a moving Gaussian source over a 0.4 mm Ti-6Al-4V powder layer at 30 W absorbed power and 800 mm/s, with conduction, convection, and radiation boundary conditions, and localized mesh refinement (40 µm elements) along the scan track.
- **Scan-strategy result:** a linear pass peaked at ~2264 K, while a full circular lap reached ~2405 K - ~475 K above liquidus - showing heat accumulation from a revisited track rather than from higher input power.

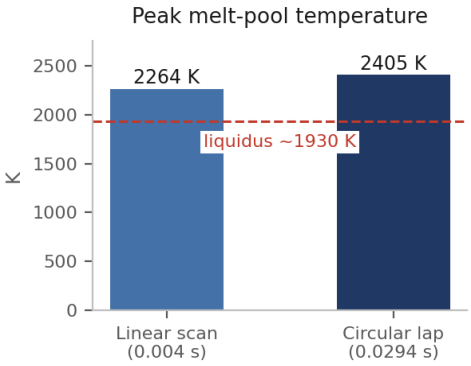


Fig. 4 - Peak melt-pool temperature by scan strategy

CAD Modeling - Personal Projects

Blender Gear - Part Modeling

Personal project | SolidWorks

Modeled the part in SolidWorks as a single solid body, adding and removing material feature by feature.

- **Feature tree:** 10 features - four boss extrudes, four cut extrudes, and two chamfers.
- **Geometry:** cylindrical body with raised teeth and a central cutout.

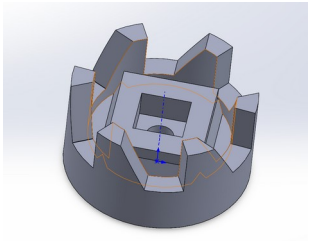


Fig. 5 - Blender Gear model (SolidWorks)

Lofted Earbud Model

Personal project | SolidWorks

Modeled the part in SolidWorks as a single solid body using extrudes, cuts, and lofts.

- **Feature tree:** 16 features - two boss extrudes, 11 cut extrudes (one thin-feature cut), and three lofts, plus one reference plane.
- **Surface blending:** used lofts and an added reference plane to blend between profiles.

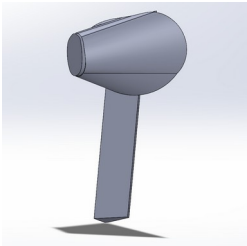


Fig. 6 - Earbuds model (SolidWorks)

Teaching Experience

Graduate Teaching Assistant - Manufacturing Processes (MEE 351) & Modern Manufacturing Methods (MAE 545)

Arizona State University | 100 and 70 students respectively

Graded exams and assignments held office hours, and developed course material and quizzes across two manufacturing courses.

- **Process improvement, applied to grading:** redesigned the quiz and attendance format and rebuilt the assignment grading method to produce consistent, equitable outcomes across a combined 170 students.

Skills & Certifications

Software

- SolidWorks, AutoCAD, Siemens NX
- COMSOL Multiphysics, ANSYS
- JMP, Excel
- Copilot Studio, SharePoint

Methods

- Time study & capacity analysis
- Design of experiments, ANOVA
- Automation ROI / NPV modeling
- Transient thermal FEA

Certifications

- Lean Six Sigma Yellow Belt - ASU, 2025
- CAD Certification - Indo-German Tool Room